Natural variation in *Arabidopsis* light/dark metabolism tests a carbon-starvation model of plant spaceflight transcriptomes

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Abstract

A gas-transport and photosynthesis model of the BRIC-LED-001 flight (OSD-522) predicted that microgravity lowers net assimilation whenever plants are lit, so that lit spaceflight transcriptomes should carry a carbon-starvation signature and dark ones should not. It also predicted metabolite shifts that could not be checked, because NASA's Open Science Data Repository holds no plant metabolome. Here we test that model against measured light/dark metabolism: leaf metabolomes of 259 accessions before and after six days of darkness, 301 accessions under control and a 32 °C darkness stress with their metabolite GWAS loci, and a light/dark time course. Darkness raised leucine and isoleucine in every accession of both natural variation panels, and the model's compound predictions agreed with the measured direction for 5 of the 6 compounds that could be compared. Glutamate, which the model groups with the rising amino acids, fell, and 41 robust darkness responses, among them TCA-cycle acids and flavin and nicotinamide cofactors, were not predicted. Replacing the model's 21-compound metabolite layer with a 96-compound measured one left the PaintOmics pathway result unchanged, with the same 6 pathways significant and identical hub statistics for shared compounds, so in this integration the gene layer decides the outcome. Of 36 pre-registered gene-set outcomes, 18 matched. None of the predicted lit-flight starvation signals reached significance, while carbon-starvation and branched-chain amino acid catabolism genes moved strongly in three dark flight contrasts, down in two BRIC canister experiments and up in the leaves of dark-grown CARA seedlings. Natural-variation GWAS gene sets showed no light-specific signal. With six plants per arm a targeted GC-MS panel could detect only changes of about 1.48-fold or larger; the model's predicted 0.13 $\log_2$ shifts would need 92 to 1,342 plants per arm. The carbon-starvation signal in plant spaceflight transcriptomes is real, but it is not specific to light, and its sign differs between flight experiments.

Keywords: spaceflight, *Arabidopsis*, natural variation, carbon starvation, darkness, metabolomics, pre-registration

1 Introduction

A companion study [1] coupled a lattice-Boltzmann model of leaf boundary-layer gas transport to a Farquhar–von Caemmerer–Berry model of photosynthesis and asked what the sealed, illuminated BRIC-LED canister of OSD-522 [2] does to the CO_2 reaching Rubisco. It predicted that microgravity, by suppressing buoyant convection, leaves a persistent deficit in net assimilation whenever the plants photosynthesise, and that the flight transcriptome should therefore show a mild carbon-starvation response. Because there is no assimilation to lose in the dark, the same model predicts no such response in dark-grown flight experiments.

That study left four things open. Its metabolite predictions could not be tested, because the NASA Open Science Data Repository [3] contains no plant metabolome; it submitted them to PaintOmics [4] as a predicted third layer instead. Its "starvation amino acids" (asparagine, valine, leucine, isoleucine, glutamine and glutamate) were given one shared fold change with a gain fixed at 1, never fitted. Its gene-set tests were Mann–Whitney comparisons, which ignore the correlation between genes. And across six flight contrasts the carbon-starvation marker genes moved significantly in dark experiments, where the model predicts nothing, an observation it recorded as unexplained.

Natural variation offers a measured answer to the first two questions and a new test of the third. Zhu and colleagues profiled the leaf metabolome of natural accessions before and after six days of extended darkness [5, 6]; 259 have both timepoints in the published per-accession values. Wu and colleagues profiled accessions under control conditions and under a stress the supplement describes as darkness at 32 °C, and mapped metabolite GWAS loci in each [7]; 301 have paired values here. Their supplement also includes LC-MS profiles from the Caldana time course, in which plants grown at 21 °C in the light were kept there or moved to darkness [8]. Together they say how carbon starvation actually moves *Arabidopsis* leaf metabolites, how consistently across genotypes, and which loci shape that response in light and in darkness.

We use them in four ways: to score the model's compound predictions; to replace its predicted metabolite layer with a measured one and resubmit to PaintOmics; to test the model's light/dark prediction on six flight contrasts with gene sets built from that natural variation, under nulls that respect gene correlation and with predictions committed before any test ran; and to ask how many plants a targeted metabolite panel on flight material would need.

2 Results

2.1 A measured light/dark reference

From the three sources we built a per-compound reference (Methods). Zhu's and Wu's responses were taken per accession, as 6 d minus 0 d of darkness and as stress minus control respectively, and a source counted for a compound when at least 75 % of accessions moved the same way; the Caldana response, 21 °C dark minus light at the timepoints from 6 h, counted at Welch $p < 0.05$. Of 161 source-compound responses, 123 could be assigned a KEGG compound through the KEGG REST API [9, 10], giving 96 compounds. Of these, 45 had a robust consensus (at least two sources agreeing, or one with at least 90 % of accessions), 22 a single qualifying source, 28 none and 1 conflicting sources; 24 rose in darkness and 43 fell.

The branched-chain amino acids dominate. Leucine rose by 4.92 $\log_2$ units after six days of darkness and by 4.74 under the darkness stress, in every accession of both panels, and by 2.40 in the Caldana time course; isoleucine rose by 3.15 and 4.42. This is the response Caldana et al. attributed to protein degradation followed by amino acid catabolism as the main energy supply once photosynthesis stops [8].

2.2 The model's compound predictions against measured darkness

Under a rule fixed before the comparison was run (the reference itself had been inspected; Methods), we compared the 12 predictions the model derives from a carbon deficit, its photosynthate and starvation blocks, with the reference (Fig. 1a). Its photorespiratory (C_2 cycle) and RuBP predictions were not scored, because darkness stops carboxylation and oxygenation together and so cannot test them.

Of these, 5 agreed in direction: maltose fell, and asparagine, valine, leucine and isoleucine rose. One disagreed. Glutamate, which the model raises with the other starvation amino acids, fell under the darkness stress (-0.50 $\log_2$, 97 % of accessions). Glutamine, also predicted to rise, did not move (-0.02, 50 % of accessions in either direction), and the reference held no light/dark measurement for the other five photosynthate compounds (3-phosphoglycerate, sucrose, glucose, fructose and starch), so 6 predictions remain untested. Magnitudes were far from uniform: the model gives all six starvation amino acids the same fold change, but leucine's standardised darkness response was 2.8 times valine's and 2.7 times asparagine's.

The reference also shows what the model leaves out (Fig. 1b): it did not predict 41 robust darkness responses. They include malate (-1.19 $\log_2$) and 2-oxoglutarate (-1.68), both TCA-cycle intermediates that the model excluded because their direction under carbon limitation seemed uncertain; the reference says both fall. Also missing were the flavin and nicotinamide cofactors (FAD, FMN, riboflavin, NAD^+), adenosine and guanosine, glutathione and ascorbate, all of which fell, and aromatic amino acids, arginine, allantoate and purine, which rose.

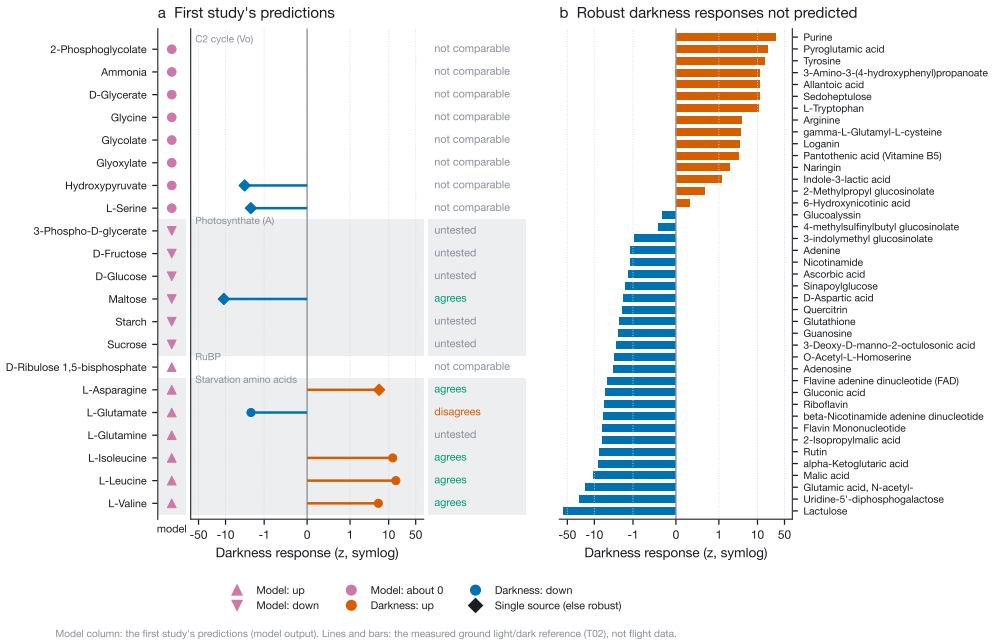


Fig. 1 The first study's metabolite predictions set against measured darkness. a, The 21 predicted compounds, grouped by the model's driver. The model column gives the predicted direction in OSD-522 (model output); lines give the measured light/dark response as a standardised effect (T02; circles robust, diamonds single source). Verdicts follow the pre-registered rule; C_2-cycle and RuBP predictions cannot be tested by darkness. **b**, Robust darkness responses the model did not predict.

2.3 A measured metabolite layer leaves the pathway result unchanged

We replaced the first study's metabolite layer in PaintOmics with the reference: all 96 compounds, each valued at the model's own carbon-deficit magnitude ($0.13\,\log_2$) in its measured direction, with the 45 robust compounds marked as changed. The gene and protein files were byte-identical to the first study's job, and the database, organism and settings unchanged (Methods). The layer is a ground reference, not a flight measurement: it says what carbon starvation does to these compounds, which is what the model claims lit flight is.

The pathway result did not move (Fig. 2a). The same 6 of 232 pathways passed a Benjamini–Hochberg false discovery rate [11] of 0.05 in both jobs, led in both by protein processing in the endoplasmic reticulum (combined $p = 8.65 \times 10^{-9}$). The rank correlation of all combined p-values was 0.95. Glyoxylate and dicarboxylate metabolism, where the C_2 cycle sits, moved from rank 214 to 193, still far from significance.

The metabolite hubs changed only where the compound list changed (Fig. 2b). Leucine, valine, isoleucine and glutamate remained hubs, and malate, 2-oxoglutarate, glutathione, ascorbate, glucobrassicin, γ-glutamylcysteine and 2-isopropylmalate joined them (11 hubs from 32 compounds, against 7 from 15). The pre-registered

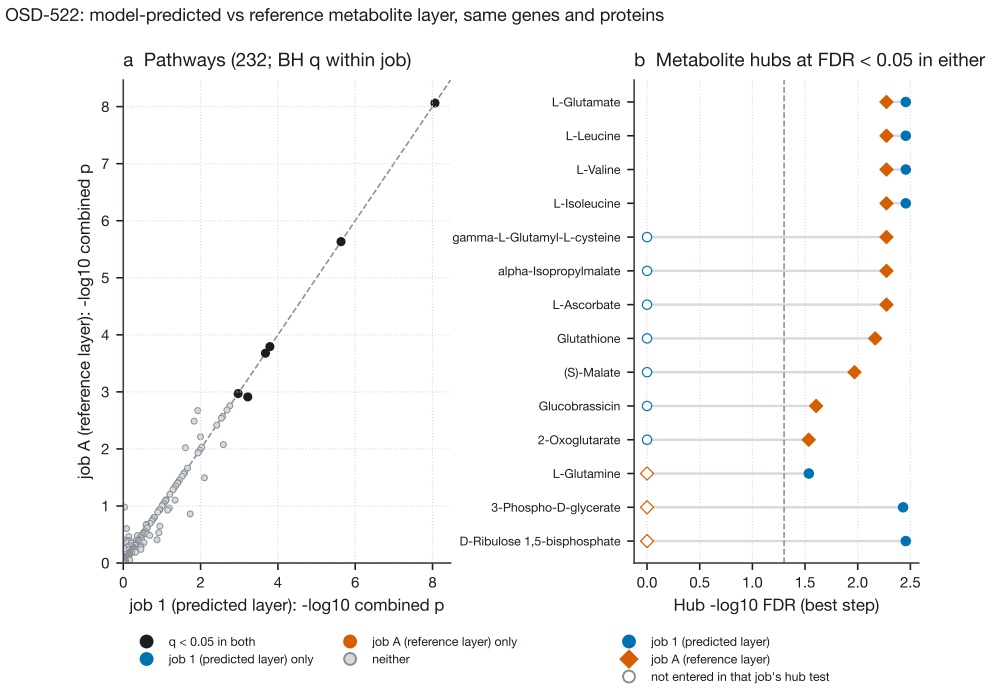


Fig. 2 Replacing the predicted metabolite layer with the measured reference. OSD-522 genes and proteins are identical in both jobs. **a**, Combined Fisher p for all 232 pathways; colours mark BH $q < 0.05$. **b**, Metabolite hubs at FDR < 0.05 in either job; open symbols mark compounds not entered in that job's hub test.

check held: for every compound and step the two jobs share, the raw hub p-value was identical (16 compound-steps). A compound's hub status is set by the differentially expressed genes around it in the KEGG network, not by the value it is given, so what the metabolite layer contributes to this integration is its compound list.

2.4 Light and dark flight in the same hardware

OSD-678 [12] grew Col-0 seedlings in the same CARA plates on the same flight, either in the ambient light of the station's cabin wall (4–6 μmol m^{-2} s^{-1}) or wrapped in black-out cloth from germination, which makes it the one place where light can be compared without changing hardware, genotype or mission. We submitted both of its gene layers with the same reference metabolite layer (Fig. 3). The light contrast had more differentially expressed genes than the dark one (7,790 against 2,112 at an adjusted $p < 0.05$) and more significant pathways (24 against 10); the two contrasts' pathway rankings were only moderately related (rank correlation 0.51). Photosynthesis was the most significant pathway in the dark contrast ($p = 1.03 \times 10^{-12}$) and did not pass correction in the light one ($q = 0.131$). Valine, leucine and isoleucine degradation was nominally significant in the dark ($p = 0.00506$) and not in the light ($p = 0.0854$).

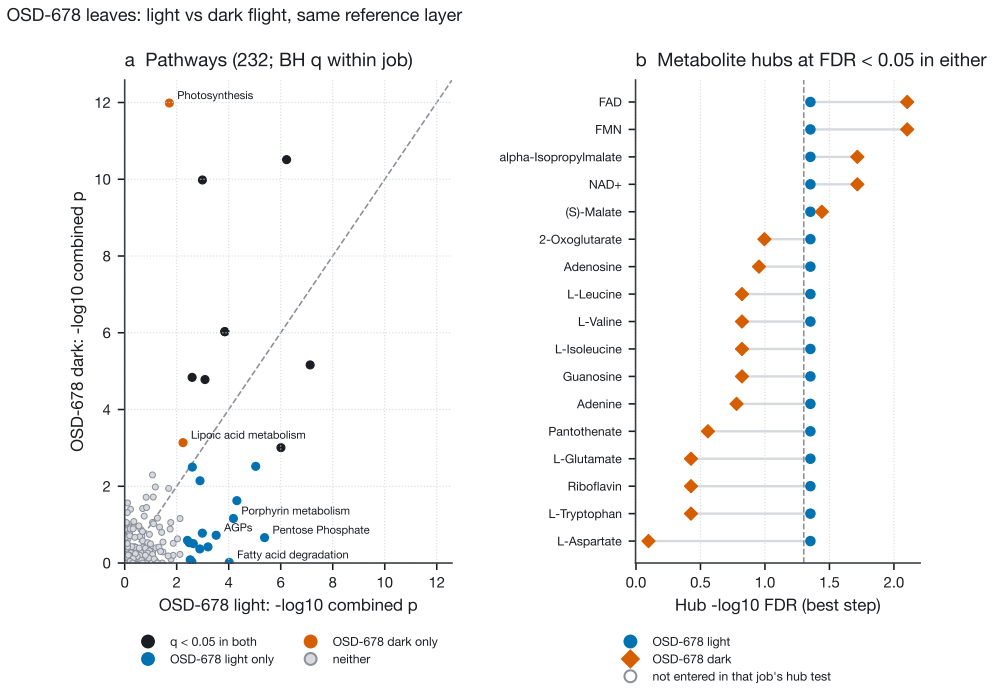


Fig. 3 OSD-678 light and dark flight contrasts with the same reference layer. a, Combined Fisher p for all pathways; the four most separated single-contrast pathways on each side are named. **b**, Metabolite hubs at FDR < 0.05 in either contrast.

In the light contrast 17 metabolite hubs reached FDR < 0.05, against 5 in the dark. The branched-chain amino acids, glutamate and 2-oxoglutarate were hubs only in the light, which is the direction the model expects, but every light hub sat at the same adjusted value (0.044) just inside the threshold, and the light contrast had 3.7 times as many differentially expressed genes to place around them. The five shared hubs (malate, FAD, FMN, NAD^+ and 2-isopropylmalate) were stronger in the dark.

2.5 Pre-registered light/dark gene-set tests

Before any gene-set statistic was computed on a flight contrast, we committed six gene sets, the predicted outcome for each in lit and dark flight contrasts, and every test setting (prereg/PREDICTIONS.md; Table 1). Two sets are signed: the first study's carbon-starvation (DIN) set, re-tested here, and KEGG's valine, leucine and isoleucine degradation pathway (53 genes), the catabolic route of the strongest darkness response. Four come from the Wu GWAS: genes in loci found only under control, only under the darkness stress, under both, and in the loci of metabolites with a directional darkness response. For OSD-522 the null was every one of the 924 ways to relabel its twelve samples. The other five contrasts were tested on GeneLab's published statistics, with too few replicates for a useful sample permutation (three per arm in OSD-38 and OSD-678) or samples that could not be assigned to an arm from their names, so their

Table 1 Pre-registered gene sets and predictions. Statistic: signed sets test the mean gene statistic, unsigned sets the mean absolute statistic, each against the rest of the genome. Lit contrasts: OSD-522, OSD-678 light, OSD-427. Dark: OSD-38, OSD-321, OSD-678 dark.

Set	Genes	Definition	Statistic	Lit	Dark
DIN (first study)	5	KEGG descriptions matching DINn, the first study's rule	signed	up	none
BCAA degradation	53	KEGG ath00280	signed	up	none
Wu stress-only	491	GWAS regions found only under 32 °C darkness	unsigned	enriched	none
Wu darkness metabolites	540	Loci of metabolites with a directional darkness response	unsigned	enriched	none
Wu control-only	301	GWAS regions found only under control	unsigned	none	none
Wu shared	1,954	GWAS regions found under both	unsigned	none	none

null was a genomic rotation that moves a set along the genome and keeps its size and clustering (Methods).

Of the 36 outcomes, 18 matched (Fig. 4), but the matches are not where the model needs them. In the lit contrasts none of the predicted signals passed correction: 5 of 18 lit outcomes matched, all of them comparator sets predicted to stay flat. The DIN set leaned upward in all three lit contrasts (OSD-522 $p = 0.0281$, OSD-678 light $p = 0.089$, OSD-427 $p = 0.0318$; $q \geq= 0.127$ throughout), and so did branched-chain catabolism in OSD-678 light ($p = 0.0397$, $q = 0.143$). For OSD-522 the first study had reported the DIN set at $p = 0.0026$ by Mann–Whitney, and the same test here reproduces that value; under the sample permutation, which keeps the correlation between genes, it weakens to $p = 0.0281$.

The dark contrasts carried the strongest signals of all, where the model predicts none. The DIN set fell in both BRIC-PDFU experiments (OSD-38 [13], $q = 0.0108$; OSD-321 [14], $q = 0.0018$) and rose in the leaves of dark-grown CARA seedlings (OSD-678 dark, $q = 0.0048$). Branched-chain catabolism followed it, down in OSD-321 ($q = 0.0018$) and up in OSD-678 dark ($q = 0.009$). The natural-variation GWAS sets gave nothing light-specific: the stress-only and darkness-metabolite loci were never enriched after correction, and the one significant enrichment, of shared loci in OSD-678 light ($q = 0.0114$), was in a comparator set the model makes no claim about.

2.6 What a targeted GC-MS panel would need

To ask whether the predicted metabolite shifts could be measured on archived flight material, we estimated per-compound replicate variance from Col-0 wild-type control groups in MetaboLights GC-MS studies [15]: 19 groups in 6 studies [16–19], 29 compounds from the model's predictions and the reference (Fig. 5). The median within-group standard deviation of $\log_2$ intensity ran from 0.31 (sucrose) to 1.21 (isoleucine).

With six plants per arm, OSD-522's RNA-seq design, the smallest change detectable at 80 % power was 0.56 $\log_2$ (1.48-fold), for sucrose. The model's predicted

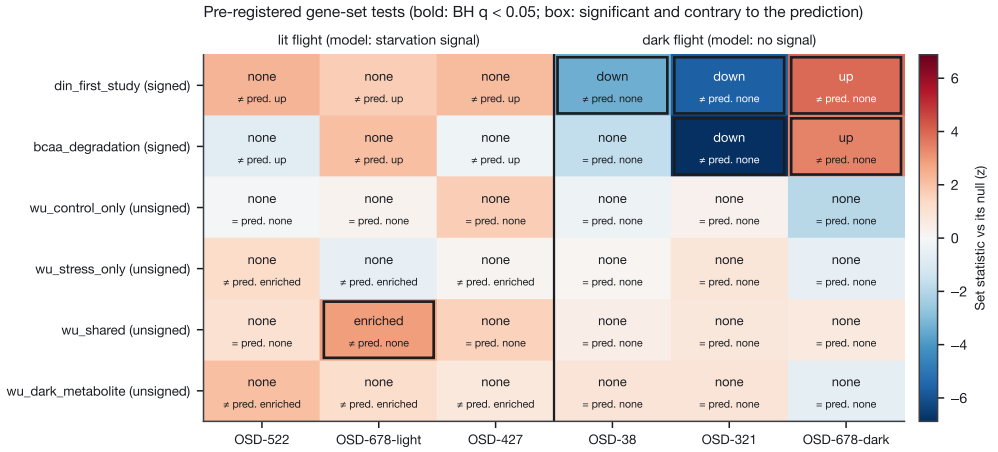


Fig. 4 Pre-registered light/dark gene-set tests. Colour: the set statistic against its null as a z-score (OSD-522, sample permutation; others, genomic rotation). Each cell gives the observed outcome (bold where BH $q < 0.05$ over the 36 tests) and whether it equals (=) or contradicts ($\neq$) the prediction; boxes mark significant contradictions.

0.13 $\log_2$ shift would need 92 to 1,342 plants per arm, and its 0.035 $\log_2$ shift at OSD-522's own growth light at least 1,293. A two-fold change, the scale of the measured darkness responses, would need at most 24.

3 Discussion

3.1 Compared with the first study

Table 2 sets the two studies side by side. The measured data support the model's metabolic picture in outline: in real carbon starvation the branched-chain amino acids and asparagine rise and maltose falls, as it predicted. They correct it in detail. Glutamate falls rather than rising, glutamine does not move, the starvation amino acids differ several-fold in how strongly they respond, and the TCA-cycle acids the model declined to predict fall consistently.

The light/dark prediction fares worse. The model's mechanism runs through photosynthesis, so its signature should appear in light and vanish in the dark. With nulls that respect gene correlation and predictions fixed in advance, no lit contrast produced a significant starvation signal, while three dark contrasts produced the strongest ones in the dataset. The first study reported that its starvation-over-photosynthesis contrast separated lit from dark experiments; a single-set test shows why that ordering is not a light effect. The starvation genes themselves move most in darkness, in opposite directions in BRIC-PDFU seedlings and in CARA leaves.

3.2 Why the dark contrasts carry the signal

We cannot say from these data, and offer possibilities to test rather than conclusions. In all three dark contrasts both arms are seedlings grown in darkness from germination [12–14], so the flight-versus-ground contrast compares two starved states, and a

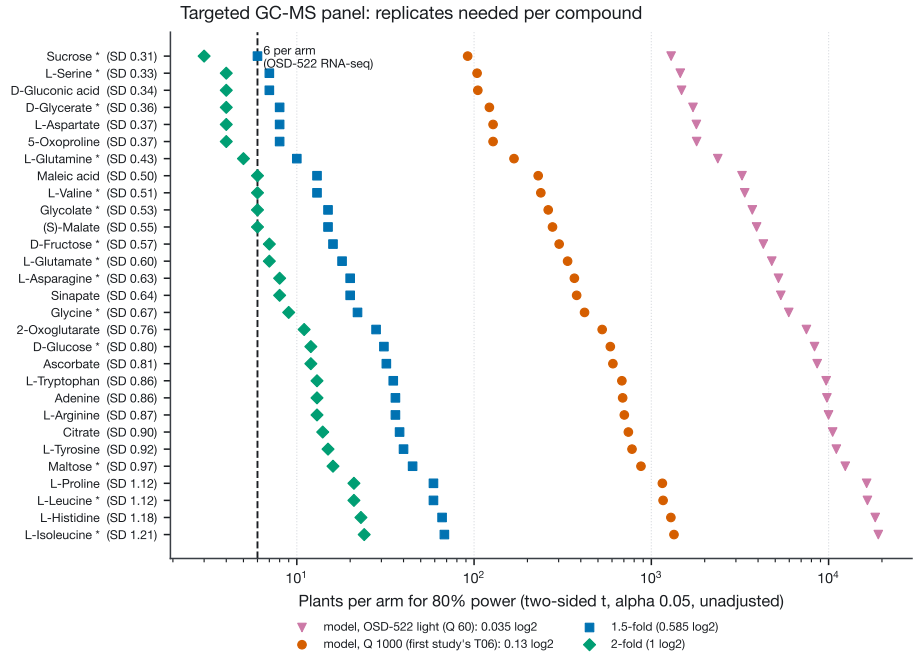


Fig. 5 Plants per arm for a targeted GC-MS panel. Two-sided two-sample t-test, 80 % power, $\alpha = 0.05$ unadjusted; variance from Col-0 control groups in MetaboLights GC-MS studies. Asterisks mark compounds the first study predicted.

difference in the starvation programme means the flight seedlings were starved more, less or differently than their ground controls. The direction is not consistent: the programme is lower in flight in the two BRIC-PDFU canister experiments and higher in the CARA plates, which differ in hardware (a sealed canister against a plate on the cabin wall) and in what was sequenced (whole seedlings against dissected leaves). Either way, a carbon-starvation programme responds to spaceflight conditions that the boundary-layer mechanism cannot reach, because in the dark there is no assimilation for it to reduce.

3.3 Limitations

The reference is ground darkness, a far larger perturbation than a 9 % assimilation deficit, so it fixes directions rather than flight magnitudes, and its use as a stand-in for flight rests on the first study's hypothesis. The Caldana genotype is not stated in the sources we could read (TODO: confirm from the paper), and the Wu stress condition is taken from the supplement. Gene-set tests on GWAS regions include many genes that do not cause the association, which dilutes any signal. The five contrasts other than OSD-522 were tested with a rotation null, which is weaker than a sample permutation; OSD-38 and OSD-678 have three replicates per arm. OSD-678's light arm was the cabin's ambient light of 4–6 μmol m^{-2} s^{-1} [12], well below OSD-522's 60, where

Table 2 The first study and this one, claim by claim.

Claim	First study [1]	This study
Starvation amino acids rise together	Predicted together, one shared gain of 1, unfitted	Leu, Ile, Val, Asn rise; Glu falls; Gln flat; Leu responds 2.7–2.8 times as strongly as Val or Asn
TCA-cycle acids	Not predicted (direction uncertain)	Malate and 2-oxoglutarate fall robustly, each in two sources
Metabolite layer in PaintOmics	21 predicted compounds	96 measured compounds; pathways unchanged; hub p identical for shared compounds
Lit flight shows a starvation signal	DIN up in OSD-522 ($p = 0.0026$, Mann–Whitney)	Leans up in all three lit contrasts; none significant after correction under gene-correlation-aware nulls
Dark flight shows none	Predicted none; dark inversions unexplained	Strongest signals are in dark-grown seedlings; sign differs between BRIC canisters and CARA plates
Metabolite shifts are measurable	Proposed a 20–30 compound panel on archived material	Needs 92–1,342 plants per arm for 0.13 $\log_2$

the model's deficit is smaller still, so its light/dark comparison is a weak test of the model's light dependence. PaintOmics hub counts rise with the number of differentially expressed genes, which differs 3.7-fold between the two OSD-678 contrasts. The GC-MS variance pools leaf and shoot studies from different laboratories and timepoints. And the pre-registration was written by the same analyst on the same day as the tests, after the reference had been built; its value lies in the commit order, which shows that no prediction or setting was changed after a flight result was seen.

4 Methods

Inputs and provenance. Every input is vendored with its source and SHA-256 in `data/*/PROVENANCE.md`. Derived natural-variation tables (Zhu per-accession darkness responses, Wu leaf metabolomes with one ion mode per metabolite, Wu GWAS loci and condition regions) and the OSD-522 raw counts are read from the Arabidopsis Atlas at commit 69b21c6 [20]. The first study's PaintOmics upload, job record, predictions and study registry are read from its repository at commit 515226d [1]. Per-contrast GeneLab differential-expression statistics for OSD-38, OSD-321, OSD-678 light and dark and OSD-427 [12–14, 21] keep only the matched flight-versus-ground columns named in that registry. The Wu supplementary tables 7 and 8 are fetched to a git-ignored cache and only derived values are committed.

Light/dark reference. Zhu: per-accession 6 d minus 0 d. Wu: stress minus control per accession, one ion mode per metabolite, metabolites without a paired value excluded. Caldana (Wu supplementary table 7): mean 21 °C dark minus 21 °C light over the timepoints at or after 360 min, three replicates each (14–18 values per compound), Welch t-test, peaks restricted to those Wu named with a KEGG identifier. A Zhu or Wu response qualified when its median was non-zero, at least 75 % of at least 20

accessions moved in its direction; a Caldana response qualified at $p < 0.05$. Consensus: conflicting if qualifying sources disagreed, robust if at least two agreed or one had at least 90 % of accessions, single otherwise. The reference value is the mean standardised effect of the qualifying sources (response divided by the source's SD across accessions or light replicates). Compound names were matched to KEGG by the REST API, a name counting only when it equals a KEGG synonym after case, punctuation and an L-/D- prefix are ignored; every candidate is recorded and every correction is listed with the KEGG name that confirms it (`data/reference/kegg_overrides.tsv`). Amino acids take their L-forms, since LC-MS does not resolve enantiomers.

Pre-registration. `scripts/11_prereg.py` wrote the gene sets, genome order, settings and predictions to `prereg/`, committed alone before any test script existed; the test script reads those files and will not run without them. The DIN set was flagged as a re-test, since the first study had already reported it in all six contrasts.

Gene-set tests. OSD-522 Wald statistics were recomputed from raw counts with PyDESeq2 [22, 23], six flight against six ground shoots, genes with a total count of at least 10; the $\log_2$ fold changes match the first study's upload to within 10^{-6}. Its permutation null used a Welch t on $\log_2$(median-of-ratios normalised count + 1) under all 924 labellings, the observed labelling included. The rotation null shifted each set's positions in the KEGG genome order by 10,000 random offsets (seed 20261030). Two-sided p-values for signed sets, upper-tail for unsigned; BH correction over the 36 primary tests. Before testing, the script plants a shift into 60 genes and leaves 60 control genes untouched, and stops unless both nulls find the first and not the second.

PaintOmics. Jobs were submitted to paintomics.org with organism `ath`, KEGG and MapMan, and AI interpretation off, by the multipart request the web form makes; the matching step was run in a browser, and the full job records were recovered and vendored. On each compound card, matches were reduced to the intended KEGG identifier (PaintOmics also selected CMP-KDO, neoglucobrassicin, sugar anomers and a non-KEGG $NADP^+$ identifier); one residual, (S)-allantoin on a compound of value 0, is recorded. Job identifiers: first study oK2h66740Q; A 6kBOS7AN5K; B JfVZ1L37PY; C wgT3IPi7U2.

Power. Col-0 control groups: the 0 mM salt group of MTBLS22, 41 and 42; Col-0 well-watered shoots of MTBLS2289 per timepoint; the wild-type group of MTBLS4066; the Col-0 time course of MTBLS363 per timepoint. MTBLS8105 was excluded because its values carry commas whose decimal convention cannot be determined, and the single-leaf series MTBLS7, 11, 12 and 13 because leaf position is not recorded. Compounds were matched by KEGG's ChEBI conversion or by KEGG synonym. Per compound, the median over groups of the within-group SD of $\log_2$ intensity (zeros missing, at least three values) set the required n per arm for a two-sided two-sample t-test by the noncentral t distribution.

Reproducibility. `run_all.sh` regenerates every table, figure and the numbers in this text from the vendored inputs; the PaintOmics step reads the vendored job records.

Data availability. All derived data are in the repository (https://github.com/dr-richard-barker/spaceflight-metabolism-natural-variation). [TODO — Zenodo DOI.] Source data: NASA OSDR (OSD-522, OSD-38, OSD-321, OSD-678, OSD-427),

the Zhu and Wu natural-variation datasets as cited, and MetaboLights MTBLS22, 41, 42, 363, 2289 and 4066.

Code availability. Scripts are in the same repository under the MIT licence.

Use of AI tools. [TODO — the analysis code and the first draft of this text were prepared with an AI assistant (Claude); state its use as the journal requires.]

Author contributions. [TODO]

Funding. [TODO]

Competing interests. [TODO]

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